

CODE SPORT



In this month's column, we discuss general-purpose sentence-embedding representations.

In the last two columns, we have been discussing the task of classifying insincere questions on community question answering (CQA) forums which are not true requests for information, but instead aim to convey an opinion, or try to legitimise the use of hate speech and inappropriate language. There is a recently released, publicly available dataset for this problem at <https://www.kaggle.com/c/quora-insincere-questions-classification/data>. Given a question, the task is to classify it as an insincere question or not. This is a binary classification problem. Given that CQA questions are typically short, this can also be considered a short text classification problem. In this month's column, we will discuss how we can use pre-trained sentence-embedding for this classification task.

Of late, transfer learning has become a hot topic in natural language processing. Transfer learning, in lay terms, is the ability to train a model on a sufficiently large general-purpose corpus such that the model can achieve reasonable results on downstream NLP tasks with a small amount of task-specific data. For instance, think of pre-trained word embeddings such as Word2vec or Glove. The practice is to use these pre-trained word embeddings as initial parameters for any NLP classifier model, which can then be fine-tuned based on task-specific data. For instance, if you are designing a text classification model using neural networks, the first stage in your classifier is the embedding layer, which is initialised with pre-trained word embeddings. These models operate at word level transfer learning. Given that words are much smaller units of text, they cannot serve to carry the longer contextual meanings that are possible with transfer learning. Hence there has been interest in obtaining universal sentence-embeddings which can transfer well across tasks.

Sentence-embedding is a fixed-length dense vector representation of an input sentence. There are many different ways of generating sentence-embedding representations – one of the simplest is the 'bag of

words'. The idea is to take the average of all word embedding vectors for the words present in the given sentence. A more sophisticated approach would be to take a weighted linear combination of word embedding vectors of the words in the sentence, and then remove the projections of the average vector on their first singular component. This was the method proposed by Arora *et al.* in their paper in ICLR 2018, available at <https://openreview.net/pdf?id=SyK00v5xx>.

An enhanced version of the 'bag of words' sentence-embedding approach is to use concatenated p-mean embeddings. When the simple averaging of word embeddings is used, the dimension of the sentence-embedding vector is the same as the dimension of the individual word vector. This makes the dimensionality of the sentence-embedding vector quite small — around 300. On the other hand, other sentence-embedding models are typically wide with dimensions of 512, 1024 or 4092. Wide sentence-embeddings enable capturing of more context, unlike narrow 'bag of words' sentence-embeddings. General-purpose sentence-embeddings can be learnt using supervised or unsupervised or weakly supervised techniques. Skip thoughts/quick thoughts vector based sentence-embeddings are obtained using unsupervised or weakly supervised techniques. Since the data is unlabelled, these embeddings can be obtained on a large corpus.

On the other hand, sentence-embeddings can also be created using supervised techniques. The Facebook AI Research group proposed the Infsent universal sentence-embedding model. The group used the Stanford Natural Language Inference (SNLI) corpus to train the sentence-embeddings. The SNLI Corpus consists of 570K sentence pairs in the English language. Each of these sentence pairs is annotated with one of three labels – entailment, contradiction and neutral. This corpus is used for the task of Natural Language Inference (NLI), also known as the 'recognising textual entailment' task in NLP. The entire corpus is human annotated and hence is the gold standard